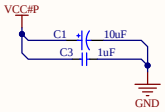
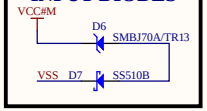
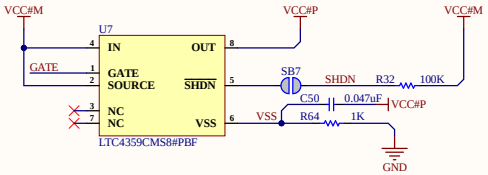
DECOUPLING CAPACITOR



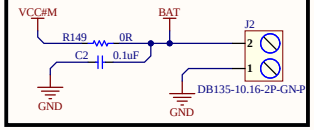
INPUT DIODES



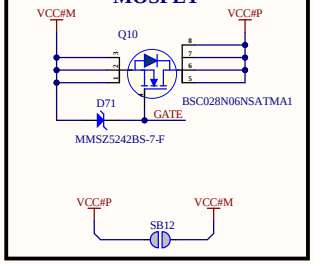
REVERSE POLARITY PROTECTION IC

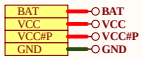


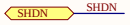
BATTERY / MAIN INPUT



MOSFET



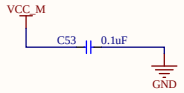




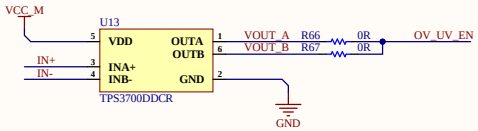
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Reverse Protection_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 11/5/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

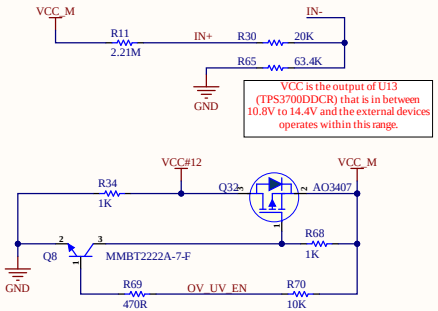
DECOUPLING CAPACITOR



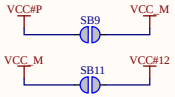
UNDER AND OVER VOLTAGE PROTECTION IC



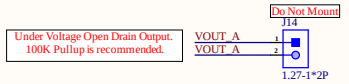
SUPPORTING CIRCUITRY



VOLTAGE TO MONITOR

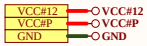


UNDER VOLTAGE OUTPUT



OVER VOLTAGE OUTPUT

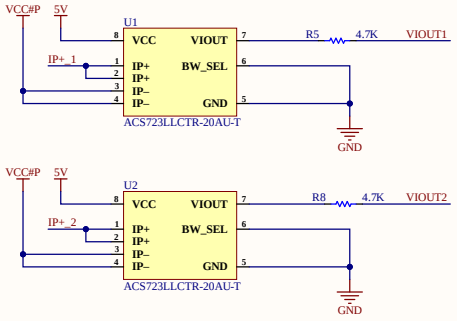




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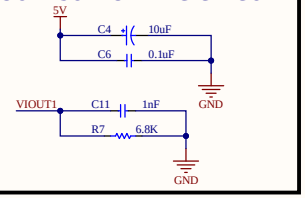
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 10/31/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

CURRENT SENSE IC

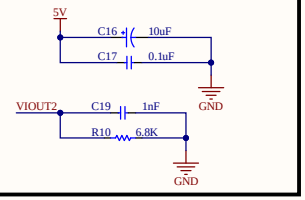


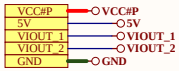
Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.

VOUT1 SUPPORTING CIRCUIT



VOUT2 SUPPORTING CIRCUIT





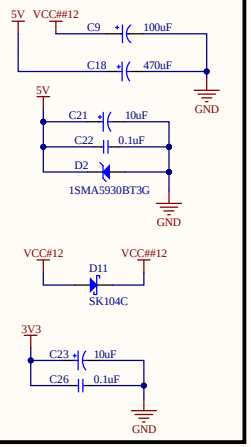




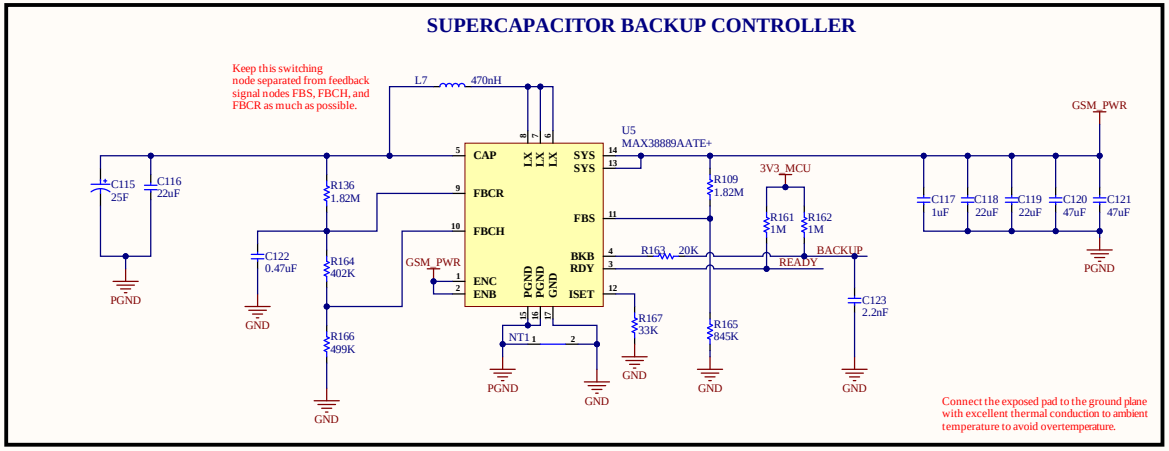
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Title: Current_Sense_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 10/31/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

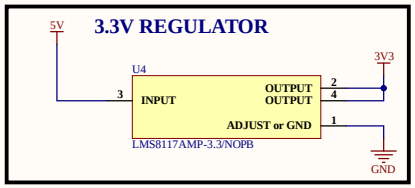
DECOUPLING CAPACITOR



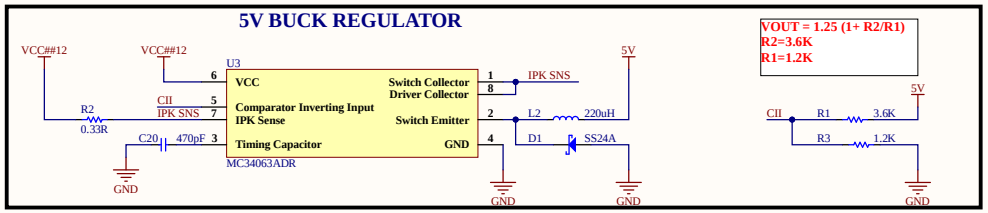
SUPERCAPACITOR BACKUP CONTROLLER



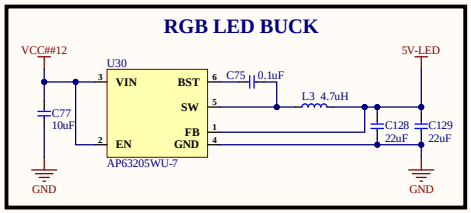
3.3V REGULATOR



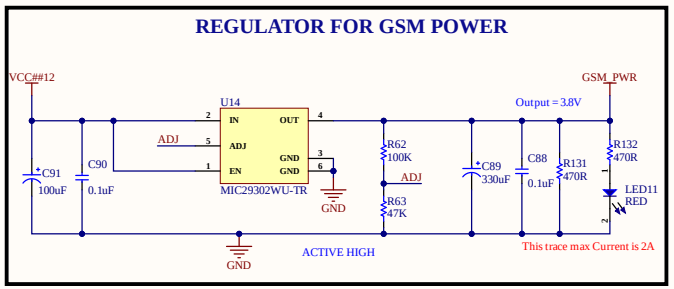
5V BUCK REGULATOR



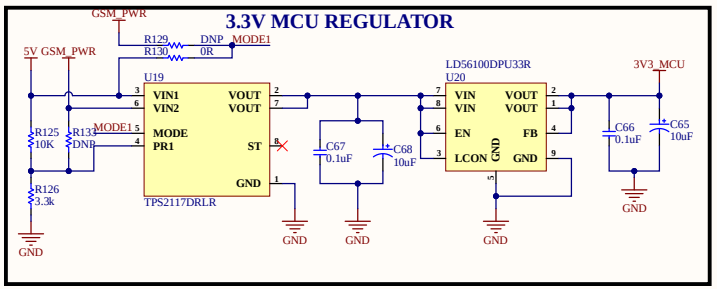
RGB LED BUCK

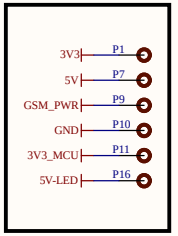


REGULATOR FOR GSM POWER



3.3V MCU REGULATOR



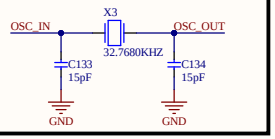


5V-LED
VCC#12
5V
GSM_PWR
3V3
3V3_MCU
GND
BACKUP
READY

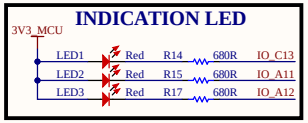
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Title: Regulator_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 11/5/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17

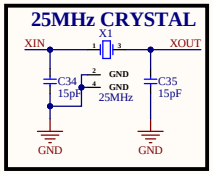
32.768KHZ CRYSTAL



INDICATION LED



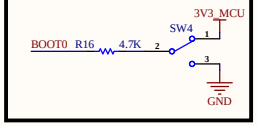
25MHz CRYSTAL



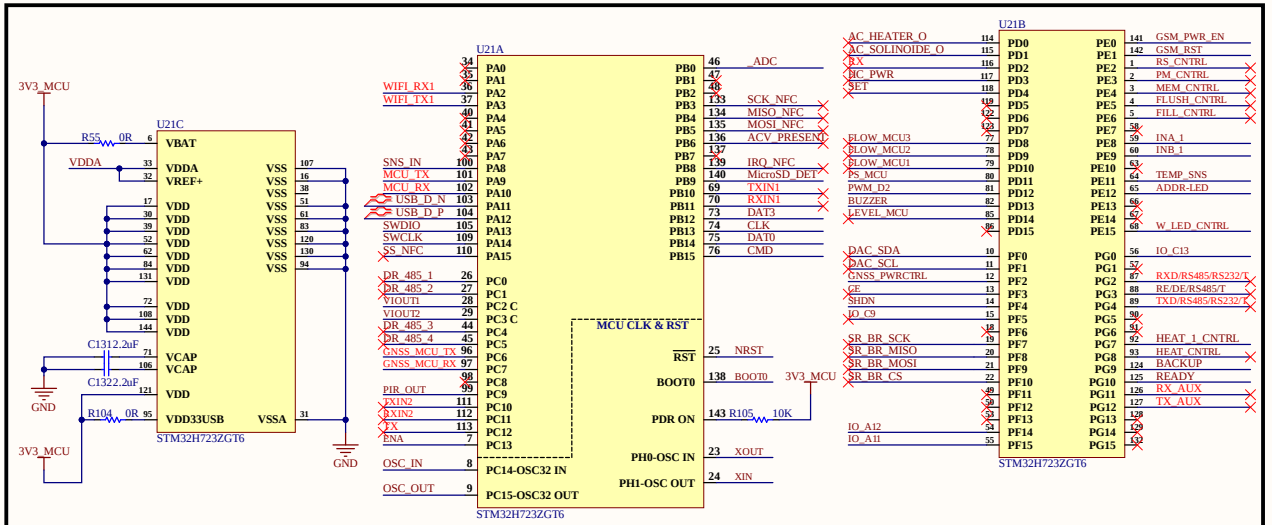
PULL UP



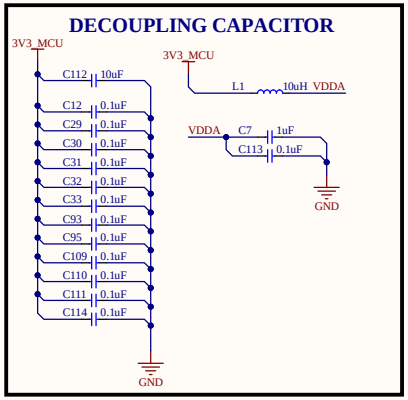
BOOT0 SWITCH



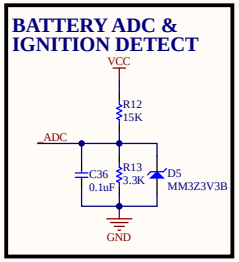
MAIN CONTROLLER



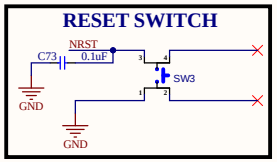
DECOUPLING CAPACITOR

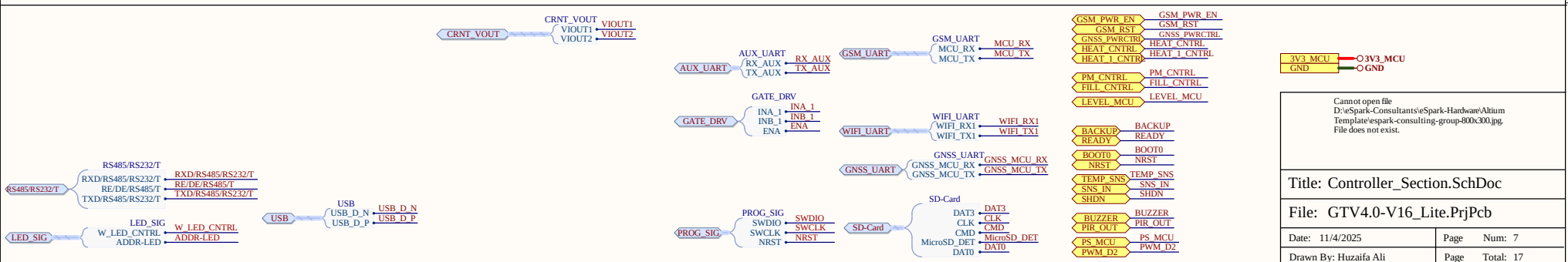


BATTERY ADC & IGNITION DETECT



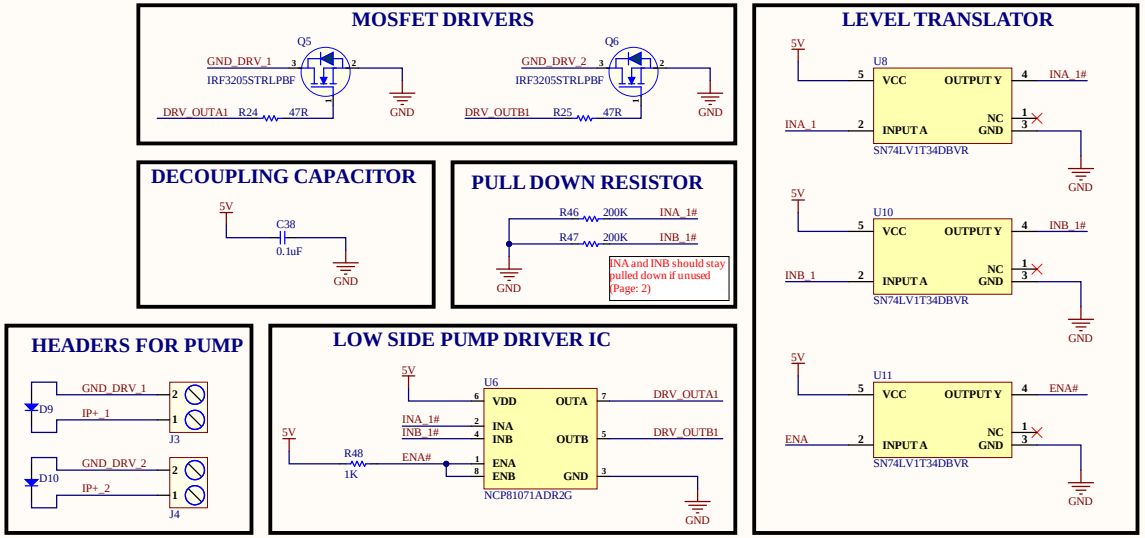
RESET SWITCH

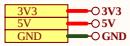




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Title: Controller_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 11/4/2025	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17





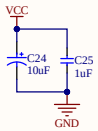




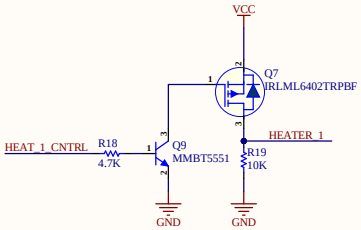
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Gate_Driver_Section.SchDoc		
File: GTV4.0-V16_Lite.PrjPcb		
Date: 11/5/2025	Page	Num: 10
Drawn By: Huzaifa Ali	Page	Total: 17

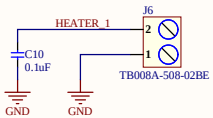
DECOUPLING CAPACITOR



MOSFET CIRCUITRY



HEATER OUT



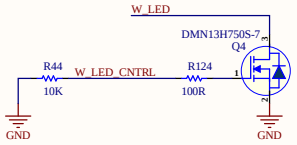




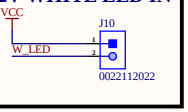
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Title: Heater_Section.SchDoc	
File: GTV4.0-V16_Lite.PrjPcb	
Date: 11/5/2025	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 17

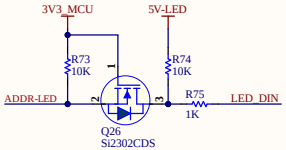
12V WHITE LED CONTROL



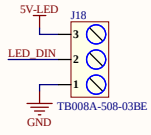
12V WHITE LED IN

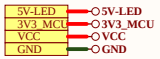


ADDRESSABLE LEDs LEVEL SHIFTER



DIAGNOSTIC LED



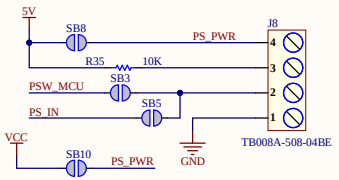




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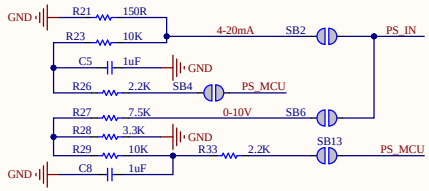
Title: LED_Section.SchDoc		
File: GTV4.0-V16_Lite.PrjPcb		
Date: 11/5/2025	Page	Num: 14
Drawn By: Huzaifa Ali	Page	Total: 17

PRESSURE SWITCH / SENSOR

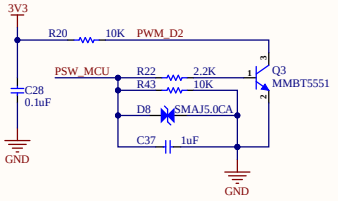


PRESSURE SENSOR

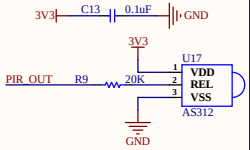
For 4-20mA respected ADC voltage will be:
4m A = 0.6V -- 20mA = 3.0V



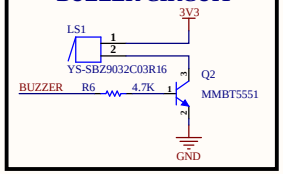
SUPPORTING CIRCUIT



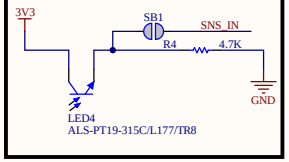
MOTION DETECTION



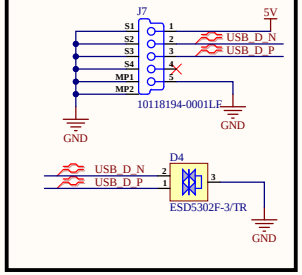
BUZZER CIRCUIT



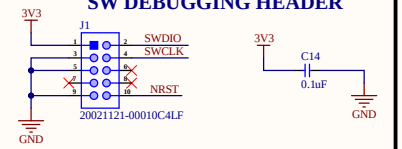
PHOTOTRANSISTOR



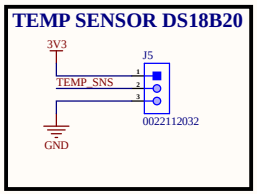
USB TYPE B CONNECTOR

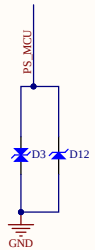


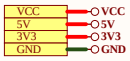
SW DEBUGGING HEADER

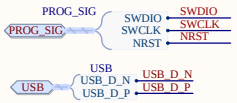


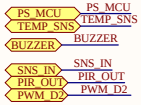
TEMP SENSOR DS18B20





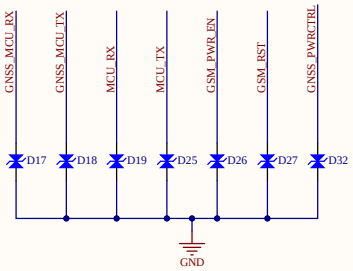


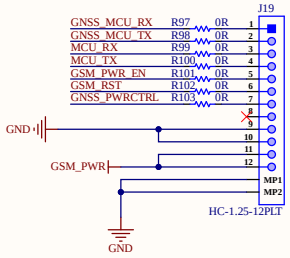




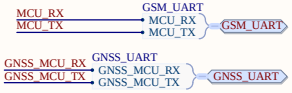
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Date: 11/5/2025	Page Num: 17
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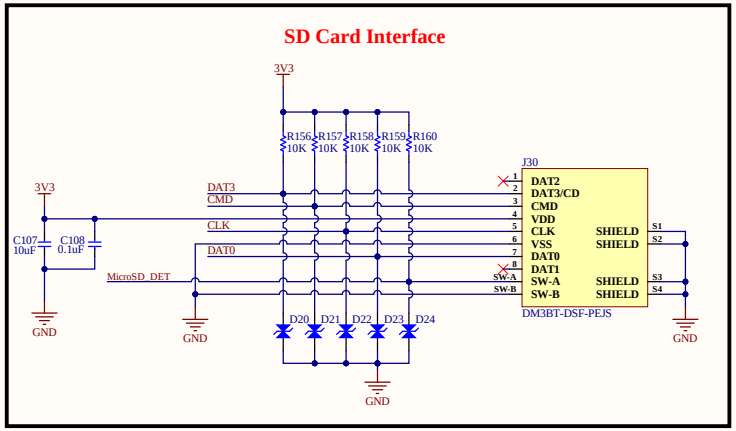






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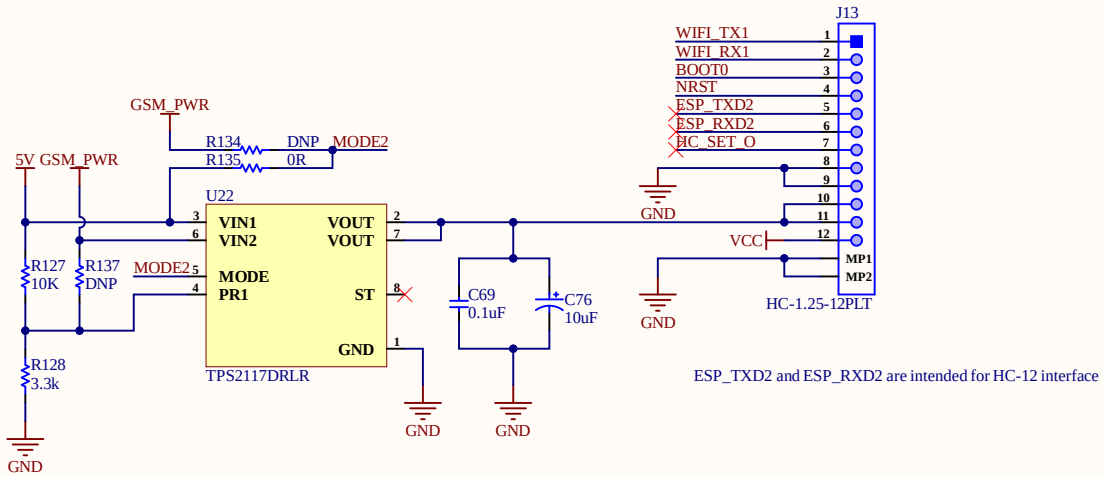


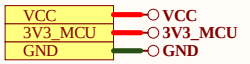
3V3
GND
3V3
GND

SD-Card
SD-Card
DAT3
CLK
CMD
MicroSD_DET
DAT0
DAT3
CLK
CMD
MicroSD_DET
DAT0

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Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.010mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 2	PP-006	0.210mm	4.4	
2	Layer 1	CF-004	0.018mm		
	Dielectric 1	FR-4	1.065mm	4.6	
3	Layer 2	CF-004	0.018mm		
	Dielectric 3	PP-006	0.210mm	4.4	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.010mm	3.5	
	Bottom Overlay				
Total board thickness:			1.611mm		

